

Physics-Informed Machine Learning for Building Energy Prediction Using Incomplete Sensor Data

Sathish Kumar Konda
25190890

Research Problem Background

Building systems are more and more utilising digital sensors to track the use of energy, temperature, humidity, ventilation, occupancy, and equipment behaviour. The sensors can be used to aid smart building management, as it allows for the facility manager to understand how the building's energy is used and how the indoor conditions change over time. The accurate prediction of building energy plays an important role in the realization of energy efficiency, reduction of energy cost, carbon-emission reduction and control of the heating, ventilation and air-conditioning system.

The data gathered from a building's sensors, however, is seldom complete, or noisy, or consistent. Sensors can stop working, go out of range, provide incorrect data or be placed in a restricted portion of a building. This can lead to machine learning models not having sufficient reliable information to make accurate energy predictions. The classic machine learning models typically rely on the patterns of past data. These models may not be very accurate if the data available includes missing measurements or is limited due to coverage by the sensors, as they don't know how the building works.

One way to overcome this challenge is through the use of physics-informed machine learning. It doesn't just use data, it uses machine learning and knowledge from the building physics. These can cover heat transfer, energy balance, occupancy effects, influence of weather, and ventilation behaviour and thermal dynamics. The model can incorporate physical knowledge into the learning process, which could enable it to make more realistic predictions in cases where the sensor data is incomplete.

This research thus aims to investigate the possibility of using physics-informed machine learning to better predict building energy use in the absence of, or with incomplete, sensor information. This uploaded topic is a unique computing research area since it touches upon topics like AI, sustainability, smart buildings, energy modelling and physics based reasoning. It is also appropriate to a computer-related project, as it can include data preprocessing, model building, evaluating the performance of machine learning techniques, and comparing them.

There are recent studies that lend credence to the significance of this matter. Ma et al. (2025) discuss physics-informed machine learning in building energy modelling, highlighting its potential to enhance the accuracy and understandability of energy prediction models. Michalakopoulos et al. (2024) also explore physics-informed neural networks to model building energy efficiency prediction. The results of these studies indicate that the topic is current and significant, and more research is required to assess the effectiveness of physics-informed models when sensors are missing.

Research Question

How does physics-informed machine learning improve the prediction of building energy, when the sensor data is incomplete, compared to traditional machine learning models?

This research question can be measured as there are clear metrics which can be used to evaluate the proposed model against traditional machine learning models. The mean absolute error, root mean squared error and coefficient of determination can be used to measure prediction performance. Model robustness can also be compared in the study when various sensor data are missing, e.g., 10%, 20% or 30% missing data. This will clarify the research question, which is focused and appropriate to being evaluated experimentally.

Justification

This is an important question to consider as building energy use is substantial and suboptimal energy use can lead to higher operating costs and environmental footprint. Building managers can make optimal decisions when energy prediction models can perform well under less-than-ideal sensor conditions. This is significant, as many real buildings lack full and high quality sensor networks.

The study is important as it tackles a real limitation of traditional machine learning. Standard models may learn properly on a complete data set, but may not be reliable if the data is missing. Physics-informed machine learning could alleviate this limitation by informing the prediction with known physics. For instance weather, occupancy, insulation and equipment usage and indoor temperature all affect energy use. A model which knows about these relationships can forecast energy demand more accurately than a model which only learns statistical patterns.

Additionally, the research adds to the knowledge by comparing and contrasting data-driven and physics-informed approaches in the context of incomplete data. This study concentrates on the problem of missing or partial sensor readings that is a realistic problem in smart buildings, while many studies discuss energy prediction. The findings would indicate if physics-informed machine learning models are more reliable, accurate, and interpretable than traditional models.

The study can be implemented on existing public building energy data or simulated building sensor data, making it feasible. A physics-informed model can be compared with a conventional model, e.g., random forest, support vector regression or a neural network. The data can be deliberately missing, to assess the performance of each model with varying proportions of sensors missing.

The study is ethical because it does not ask for the personal information of a building's users. Occupancy-related data should be anonymised and responsibly used, if applicable. Furthermore, the research is also relevant to ethical computing as it is focused on improving energy efficiency and sustainability.

Specific Items to Be Addressed

Building energy prediction will be reviewed in the first step of research, and the reason for accurate prediction of energy consumption is explained for building management. It will then explore the issue of missing data from the sensors and how this impacts on machine learning performance.

The study will also examine physics-informed machine learning, and the physics-based principles such as heat transfer, energy balance, and occupancy that can affect these will be discussed in relation to prediction models. Then a standard machine learning model will be contrasted with a physics-informed model with the same data.

The evaluation will be based on the accuracy of prediction, robustness to missing data and practical usefulness. Lastly, limitations, such as limitation of data sets, the complexity of the models, and the implementation of the sensor assumptions as well as the difference between experimental and real building deployment, will be discussed.

References

- [1] Ma, Z., Jiang, G., Hu, Y. and Chen, J. (2025) ‘A review of physics-informed machine learning for building energy modeling’, *Applied Energy*, 381, Article 125169. Available at: <https://doi.org/10.1016/j.apenergy.2024.125169>
- [2] Michalakopoulos, V., Pelekis, S., Kormpakis, G., Karakolis, V., Mouzakis, S. and Askounis, D. (2024) ‘Data-driven building energy efficiency prediction using physics-informed neural networks’, *2024 IEEE Conference on Technologies for Sustainability (SusTech)*. Available at: <https://doi.org/10.1109/SusTech60925.2024.10553513>